

Feature Selection in Statistical Machine Learning

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Lecture 4

- Recapping two definitions/perspectives of feature selection
- Re-visiting linear model example and intuition
- Feature-wise conditional independence approach
- Introducing classes of methods, filter, wrapper and embedded
- Filter methods: correlation screening in detail

Mathematical formulation: Loss function/prediction perspective

In the *population* or *ideal* setting where there is infinite data:

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = f(X_1, X_2, \dots, X_p)$$

$$S^* \in \arg \min_{S \subseteq \{1, 2, \dots, p\}, |S| \leq s} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

Population and finite sample

General feature selection:

$$S^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, |S| \leq s} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

Empirical (estimated quantities):

$$\widehat{S}^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, |S| \leq s} \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(Y^{(i)}, f(X_S^{(i)}))$$

Recall: Conditional independence perspective for feature selection

Subset of features determined as:

$$S^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, Y \perp X_{S^c} | X_S} |S|$$

We can also consider this feature-by-feature:

$$Y \perp X_j | X_{-j}, \text{ for all } j \in S^c$$

$$Y \not\perp X_j | X_{-j}, \text{ for all } j \in S$$

Linear model/housing data example

Consider the following linear model (X_1, X_2, X_3, Y) :

$$\begin{aligned} Y &= X_1 + X_3 + \epsilon_Y \\ X_2 &= \frac{1}{2}X_1 + \epsilon_2 \end{aligned}$$

$(X_1, \epsilon_2, X_3, \epsilon_Y)$ are all *independent* zero-mean random variables.

Question: Which features (X_1, X_2, X_3) is Y dependent on?

Question: Which features (X_1, X_2, X_3) is Y *conditionally* dependent on given all the other features?

Intuition (graphical models at high level)

$$\begin{aligned}Y &= X_1 + X_3 + \epsilon_Y \\X_2 &= \frac{1}{2}X_1 + \epsilon_2\end{aligned}$$

Since Y is a function of (X_1, X_3) , directed graph has arrows:

$$(X_1, Y), (X_3, Y)$$

Since X_2 is a function of X_1 , there is a directed arrow:

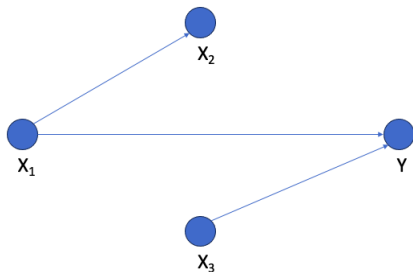
$$(X_1, X_2)$$

So overall directed graph with edges:

$$E = \{(X_1, X_2), (X_1, Y), (X_3, Y)\}.$$

Intuition (graphical models at high level)

$$E = \{(X_1, X_2), (X_1, Y), (X_3, Y)\}.$$



Question: Which features (X_1, X_2, X_3) is Y dependent on?

Which features have a *path* to Y (captures direct and indirect effects)?

Question: Which features (X_1, X_2, X_3) is Y *conditionally* dependent on given all the other features?

Which features *directly* point to Y (captures direct effects only)?

Decomposing subset selection into individual feature selection

Note that subset selection has the issue that there are 2^p subsets of features in total

Original subset of features determined as:

$$S^* = \arg \min_{S \subseteq \{1, 2, \dots, p\}, Y \perp X_{S^c} | X_S} |S|$$

There is a slightly weaker *local characterization*: Define the subset $\tilde{S}$

$$\begin{aligned} Y &\perp X_j | X_{-j}, \text{ for all } j \in \tilde{S}^c \\ Y &\not\perp X_j | X_{-j}, \text{ for all } j \in \tilde{S} \end{aligned}$$

Under some non-redundancy conditions $S^* = \tilde{S}$.

This local characterization $\tilde{S}$ is useful (especially when we later consider wrapper methods) because it means we can in theory address the problem feature by feature (only p subsets)

$\tilde{S}$ for housing data/linear model example

$$Y \perp X_j | X_{-j}, \text{ for all } j \in \tilde{S}^c$$

$$Y \not\perp X_j | X_{-j}, \text{ for all } j \in \tilde{S}$$

$Y \not\perp X_1 | \{X_2, X_3\}$ (select X_1)

$Y \perp X_2 | \{X_1, X_3\}$ (don't select X_2)

$Y \not\perp X_3 | \{X_1, X_2\}$ (select X_3)

$\tilde{S} = \{1, 3\}$ and $\tilde{S} = S^*$

Now let's talk about feature selection methods

For any feature selection method:

- Two central questions/challenges:
 - ▶ **Computational:** Can we compute $\widehat{S}^*$ in reasonable time?
 - ▶ **Statistical:** How close is $\widehat{S}^*$ to S^* ?

Slightly vague but useful classification of methods

- **Filter methods:** E.g. correlation screening, mutual information
- **Embedded methods:** E.g. Lasso, Gini/Impurity index in random forests
- **Wrapper methods:** E.g. recursive feature elimination, leave-one-covariate-out (LOCO)

Note: Original characterization in Guyon and Elisseeff more qualitative and has evolved since then

Methods may be hybrid or not easily classifiable

Characterization useful but not completely mathematical/rigorous

Population and finite sample

General feature selection:

$$S^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, |S| \leq s} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

Empirical (estimated quantities):

$$\widehat{S}^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, |S| \leq s} \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(Y^{(i)}, f(X_S^{(i)}))$$

How do we determine $\mathcal{F}$

- Typically depends on problem context and data
- In this course: will focus on 3 classes:
 - ▶ Traditional parametric functions (e.g. linear regression with or without interactions, logistic regression, SVMs, e.t.c.)
 - ▶ Decision tree ensembles (e.g. random forests, boosted trees)
 - ▶ Neural network-type (e.g. LLMs, feed-forward NNs, CNNs, RNNs, e.t.c.)
- But there are others (e.g. kernel methods/splines, additive models, e.t.c.) for which many of our methods still apply
- Some feature selection methods heavily depend on the choice of $\mathcal{F}$ while others apply more generally

Filter methods

- Example: Correlation screening
- Qualitatively, methods that *do not* on prediction function f

Embedded methods

- Example: Lasso/regularized loss
- Feature selection is part of prediction function estimation
- Parameters θ define feature selection $\{\widehat{\theta}_j = 0 \iff X_j \notin \widehat{S}\}$
- Only applies to function classes $\mathcal{F}$ that has such parameters (e.g. regression, decision trees, *not* neural networks)

Lasso example:

$$\widehat{\theta} \in \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(Y^{(i)}, \sum_{j=1}^p \theta_j X_j^{(i)}) + \lambda \sum_{j=1}^p |\theta_j|$$

Wrapper methods

- Example: Recursive feature elimination
- Attempts to solve the original optimization problem directly
- Applies to any function class $\mathcal{F}$
- Often very slow and speed-ups needed to allow computation in reasonable time

Filter methods: Bike-sharing data example

- Daily counts of rental bikes in 2011 and 2012 in Capital bikeshare system
- $n = 731$, $p = 10$, $Y = \text{'cnt'}$
- Always read attached 'readme.txt'

Learn about the data

$n = 731$, $p = 10$, $Y = \text{'cnt'}$

Variable Name	Role	Type	Description	Units	Missing Values
instant	ID	Integer	record index		no
dteday	Feature	Date	date		no
season	Feature	Categorical	1:winter, 2:spring, 3:summer, 4:fall		no
yr	Feature	Categorical	year (0: 2011, 1: 2012)		no
mnth	Feature	Categorical	month (1 to 12)		no
hr	Feature	Categorical	hour (0 to 23)		no
holiday	Feature	Binary	weather day is holiday or not (extracted from http://dchr.dc.gov/page/holiday-schedule)		no
weekday	Feature	Categorical	day of the week		no
workingday	Feature	Binary	if day is neither weekend nor holiday is 1, otherwise is 0		no
weathersit	Feature	Categorical	- 1: Clear, Few clouds, Partly cloudy, Partly cloudy		no

Correlation and sample correlation

Correlation for feature X_j , ρ_j :

$$\rho_j = \frac{\text{Cov}(X_j, Y)}{\sqrt{\text{Var}(X_j)\text{Var}(Y)}} = \frac{\mathbb{E}[(X_j - \mathbb{E}[X_j])(Y - \mathbb{E}[Y])]}{\sqrt{\mathbb{E}[(X_j - \mathbb{E}[X_j])^2]\mathbb{E}[(Y - \mathbb{E}[Y])^2]}}$$

Sample correlation from data:

$$r_j = \frac{\sum_{i=1}^n (X_j^{(i)} - \bar{X}_j)(Y^{(i)} - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_j^{(i)} - \bar{X}_j)^2 \cdot \sum_{i=1}^n (Y^{(i)} - \bar{Y})^2}}$$

where $\bar{X}_j = \frac{1}{n} \sum_{i=1}^n X_j^{(i)}$ and $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y^{(i)}$

Properties of ρ_j and r_j

- $-1 \leq \rho_j, r_j \leq 1$
- Correlation invariant under linear transformation and unit-less
- If X_j and Y are independent, then $\rho_j = 0$

Example:

$$Y = aX_j + \epsilon$$

where ϵ, X_j independent

$$\rho_j = \text{Corr}(X_j, Y) = \frac{a\text{Var}(X_j)}{\sqrt{\text{Var}(X_j) \cdot (a^2\text{Var}(X_j) + \text{Var}(\epsilon^2))}}$$

Code for generating correlations

```
import pandas as pd
import numpy as np

# Load data (download from UCI if not present)
df = pd.read_csv("day.csv")
```

```
target = "cnt"

features = [
    "temp", "atemp", "hum", "windspeed",
    "season", "yr", "mnth", "weekday",
    "workingday", "weathersit"
]

X = df[features]
y = df[target]
```

```
corr = X.apply(lambda col: np.corrcoef(col, y)[0, 1])
corr = corr.abs().sort_values(ascending=False)
```

Correlation ranking

```
[4]: corr
```

```
[4]: atemp      0.631066  
     temp      0.627494  
     yr        0.566710  
     season    0.406100  
     weathersit 0.297391  
     mnth      0.279977  
     windspeed 0.234545  
     hum       0.100659  
     weekday   0.067443  
     workingday 0.061156  
     dtype: float64
```

How do we decide if the correlation is statistically significant?

We need a hypothesis testing framework, significance level α and test statistic for which we can compute significance.

Pearson correlation statistic

For each feature X_j , correlation screening implicitly tests

$$H_0^{(j)} : \rho_{X_j,Y} = 0 \quad \text{vs} \quad H_1^{(j)} : \rho_{X_j,Y} \neq 0$$

where ρ is the **population Pearson correlation**.

Test statistic

Given sample correlation r_j with n samples,

$$t_j = r_j \sqrt{\frac{n-2}{1-r_j^2}} \sim t_{n-2} \quad \text{under } H_0.$$

We can use others (e.g. Kendall)

Filter method: Correlation screening

```
[6]: from scipy.stats import pearsonr
      from statsmodels.stats.multitest import multipletests

[8]: corr_stats = []

for f in features:
    r, p = pearsonr(X[f], y)
    corr_stats.append({
        "feature": f,
        "correlation": r,
        "abs_correlation": abs(r),
        "p_value": p
    })

corr_df = pd.DataFrame(corr_stats).sort_values(
    "abs_correlation", ascending=False
)
```

Filter method: Correlation screening

```
[9]: corr_df
```

```
[9]:
```

	feature	correlation	abs_correlation	p_value
1	atemp	0.631066	0.631066	1.854504e-82
0	temp	0.627494	0.627494	2.810622e-81
5	yr	0.566710	0.566710	2.483540e-63
4	season	0.406100	0.406100	2.133997e-30
9	weathersit	-0.297391	0.297391	2.150976e-16
6	mnth	0.279977	0.279977	1.243112e-14
3	windspeed	-0.234545	0.234545	1.359959e-10
2	hum	-0.100659	0.100659	6.454143e-03
7	weekday	0.067443	0.067443	6.839081e-02
8	workingday	0.061156	0.061156	9.849496e-02

Potentially also need to deal with multiple testing

- We do a test for each feature
- p tests (a bit of a problem when we have 10 features)
- Huge problem when p is large
- Traditionally we would use Bonferroni correction (often too conservative)
- Now more standard to use Benjamini-Hochberg correction

Filter method: Correlation screening

```
[20]: rej, p_adj, _, _ = multipletests(
    corr_df["p_value"],
    alpha=0.05,
    method="fdr_bh"
)

corr_df["p_adj"] = p_adj
corr_df["significant"] = rej
corr_df
```

```
[20]:
```

	feature	correlation	abs_correlation	p_value	p_adj	significant
1	atemp	0.631066	0.631066	1.854504e-82	1.854504e-81	True
0	temp	0.627494	0.627494	2.810622e-81	1.405311e-80	True
5	yr	0.566710	0.566710	2.483540e-63	8.278466e-63	True
4	season	0.406100	0.406100	2.133997e-30	5.334992e-30	True
9	weathersit	-0.297391	0.297391	2.150976e-16	4.301952e-16	True
6	mnth	0.279977	0.279977	1.243112e-14	2.071853e-14	True
3	windspeed	-0.234545	0.234545	1.359959e-10	1.942798e-10	True
2	hum	-0.100659	0.100659	6.454143e-03	8.067679e-03	True
7	weekday	0.067443	0.067443	6.839081e-02	7.598979e-02	False
8	workingday	0.061156	0.061156	9.849496e-02	9.849496e-02	False

Conclusion

$S = \{\text{atemp, temp, yr, season, weathersit, mnth, windspeed, hum}\}$

$S^c = \text{weekday, workingday}$

- Thoughts?
- Can we provide explanations for S and S^c ?